

The History of Computers

INSTRUCTIONS:

Read this document before you watch the video. Do your best. You do not need to understand 100%. After you have read a few times, watch the video of Baidenmann-sensei reading, and read along with him.

Important computer science words have been highlighted in blue. See the Key Vocabulary section at the end.

The History of Computers

“My mother is a computer”

Today, if someone says “my mother is a computer”, you probably think that their mom is a robot. However, before the [invention](#) of modern electronic computers, the word ‘computer’ used to mean a job that a person would do. a ‘computer’ was someone who did [computation](#). Computers were usually women. Some people, such as technicians, architects, or bankers, needed to know the answers to difficult mathematical [calculations](#). They did not always have the time or skills to ‘compute’ these calculations, so they hired a ‘computer’—someone whose job it is to compute large, difficult calculations.

The First Computation Machines

For thousands of years, people have used a tool called an abacus to do difficult calculations. An abacus is a computer because it [computes](#) calculations. Perhaps the first computation machine that looks like a modern computer was invented by the French mathematician, Blaise Pascal. In 1640, Pascal made an ‘Arithmetic Machine’, which he used to count money quickly and accurately. Later, in 1670, a German scientist called Gottfried Leibnitz continued Pascal’s work and made a machine called the ‘Step Reckoner’. It could do more advanced calculations than Pascal’s Arithmetic Machine. The Step Reckoner used only two digits, 0 and 1, to do its calculations. This system, called ‘[binary](#)’, is still the system that modern computers use today.

The Analytical Engine - The First Programmable Computer

Much later, in the 1820s, a British [mathematician](#) named Charles Babbage invented a machine called ‘The Difference Engine’, which could do even more advanced computations.

Later, Babbage invented an even more advanced machine, 'The Analytical Engine', which could do more complex calculations because it had a simple kind of [memory](#). Because the Analytical Engine had memory, you could write [programs](#) for it. Because of this, the Analytical Engine is often considered to be the first real computer. It was a machine that used binary, had memory, and could be programmed. Ada Lovelace, a very skilled mathematician, made a program for the Analytical Engine. That means Ada Lovelace was the first [computer programmer](#) in history. In 1979, computer scientists named a [programming language](#) after Lovelace, called ADA.

The First Modern Computers

People in the 19th century invented some more machines that are like modern computers, such as the German [inventor](#), Konrad Zuse, who made a machine called the Z1. The Z1, and his later machines, the Z2, Z3, and Z4, used binary (1s and 0s), and could be programmed. Zuse's machines were more [efficient](#) than previous ones, but they took up more space. The Z1 was so large, it took up the entire living room at Zuse's house.

Perhaps the most important computer scientist is Alan Turing. Turing was a mathematician who started working for the British Government in 1939. During the Second World War, the German army sent messages to their soldiers with a machine called 'the Enigma Machine'. The Enigma Machine changed the messages into a secret code, so the British could not understand the messages. Turing and other mathematicians made a computer called 'Bombe' to [decode](#) and understand the secret German messages. Then, in 1943, the Germans created a new [code](#), which the British called 'Fish'. In just one year, Turing and his team made a big computer, called 'Colossus', that could understand this secret German code. Colossus was the world's first electronic computer that could read and understand programs. Colossus got its name from its size. The computer took up an entire room. Colossus could do thousands of calculations every second. Colossus could decode a message in about 2 hours. The same message would take a team of 3 mathematicians over 6 weeks to decode! A modern PC is not much faster than Colossus, but a modern PC is much, much smaller.

With Colossus, Turing helped win the war against Nazi Germany. Despite Turing's efforts, the British government did not treat him very well after the war. Turing was a gay man, and at the time, the British government did not like people who are gay, so they treated him badly. Today, however, we remember Turing as a hero, and people often call him "the father of computers" because he created the first electronic computer.

Conclusion

Computers have advanced a lot since Turing's Colossus computer. Today, there are many specialized jobs in the field of computer science. In the listening section, we will learn about some different jobs in computer science today.

KEY VOCABULARY

Here is some key vocabulary from the reading. You can change the suffix to make different forms.

Noun	Verb	Person or Machine who does it	Meaning
invention	to invent	inventor	to create something new
computation	to compute	computer	to manage data or do advanced calculations
calculation	to calculate	calculator	to do basic math or 'arithmetic'
mathematics	(to calculate)	mathematician	often shortened to 'math'

Computer Science Vocabulary

Word	Meaning
binary	Code written with only ones and zeros (1s and 0s).
code, decode	'Code' means to change something into a special or secret language. Decode means to change code into a language you can read.
efficient	To do something in the shortest amount of time by using the least amount of energy.
memory	related to the word 'remember'. Memory in a computer keeps and processes data.
program	A system of commands for a computer. Programs are also called 'applications'.
programming language	A code that allows computer programmers to write programs efficiently. Programming languages are built on top of binary.